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Summary

Software Engineer with 4 years of experience in backend development, distributed data systems, and real-time and LLM-backed services using Python, Scala, Docker, and Kubernetes. Focused on building scalable microservices with strong observability, performance and security, paired with excellent communication skills and a passion for learning new technologies.

Experience

- ELYADATA** [🌐] Aug 2025 - Present
Tunis, Tunisia
Software Engineer II
 - Developed a real-time voice communication system enabling cellular phone calls between clients and an AI-powered virtual assistant using Twilio, Langflow, and OpenAI realtime agents.
 - Designed a generic, scalable data loader framework capable of ingesting large volumes of structured and semi-structured data from cloud storage and files (e.g. S3, CSV, Parquet), validating records via configurable business rules, and orchestrating distributed processing across multiple Kubernetes pods in parallel.
 - Built end-to-end observability by exposing Prometheus metrics, building Grafana dashboards, and integrating Elastic APM tracing for real-time monitoring, performance analysis, and improved reliability of the virtual assistant.
 - Conducted performance benchmarking with Gatling/Locust to measure the system's ability to handle up to 1000 requests per second under load.
 - Monitored and profiled JVM and PostgreSQL performance with Java Flight Recorder (JFR) and `pg_stat_statements`, tracking slow queries and open database connections to surface performance hotspots.
 - Identified and resolved performance bottlenecks across key system architecture components using observability tools and monitoring insights, resulting in improved scalability and reduced response time by 50%.**Tools:** Python, FastAPI, PostgreSQL, Twilio, Langflow, OpenAI, Prometheus, Grafana, Elastic APM, JFR, `pg_stat_statements`, Gatling, Locust, Docker, Kubernetes, Helm, REST APIs, Microservices
- COGNIRA** [🌐] July 2022 - July 2025
Tunis, Tunisia
Software Engineer II
 - Designed and developed backend features for a complex promotion-planning platform, including a key client feature, "versioning at time of finalization".
 - Architected and deployed caching systems that reduced data retrieval times by 30%, significantly enhancing user experience.
 - Designed and optimized data models for PostgreSQL and Cassandra, ensuring efficient data storage, retrieval, and scalability.
 - Conducted comprehensive unit testing with 80% code coverage, ensuring high code quality and application reliability, identifying and resolving 95% of potential issues prior to production.
 - Migrated backend services build system from SBT to Bazel, improving build performance, dependency management, and CI/CD pipeline efficiency.
 - Developed PySpark migration scripts for efficient database migration and data transformation.**Tools:** Scala, PostgreSQL, Cassandra, Kubernetes, Docker, Microservices, REST APIs, Akka, Git, Bitbucket, Agile development
- TELNET Groupe** [🌐] Jan 2022 - July 2022
Sfax, Tunisia
Software Engineer
 - Led the development of a resume scanning and classification solution using OCR and AI technologies, following Agile Scrum for timely delivery and requirement alignment.
 - Built backend microservices for classification, authentication, and document management using Python, Flask, and OpenCV, securing access with JWT authorization.
 - Optimized microservice performance with multithreading, reducing document classification processing time by 30%.
 - Developed a Flutter mobile app for scanning resumes, secure authentication, and document management, integrating contour detection to improve scan accuracy.**Tools:** Python, Flask, Flutter, NLP, MySQL, JWT, OAuth 2.0, Git, GitLab

Internships

• Future Proof

June 2021 - Aug 2021

Intern

Sfax, Tunisia

- Led the study, design, and implementation of a business process for managing a real estate advertisement database, benchmarking and configuring web scraping tools and applying Flowable, BPMN and Named Entity Recognition (NER) to streamline data collection and quality control.
 - Developed backend microservices with Spring Boot integrating machine learning models to extract key features from over 1,000 textual descriptions, over a PostgreSQL data model supporting a database of more than 50,000 records.
- Tools:** Java, Spring Boot, PostgreSQL, Flowable, Microservices, Machine Learning, Agile Development, BPMN 2.0

Projects

• FINT — Personal Finance Analytics Platform

2026

Solo project — one FastAPI backend serving a React web app and an Expo Android app

- Shipped **Ask**, natural-language Q&A over the user's own financial data: an LLM writes read-only SQL, PostgreSQL computes every total, and each answer displays the query behind it — figures are correct by construction rather than hallucinated.
- Made tenant isolation **structural rather than prompt-level**, since open registration makes prompt injection a real threat: 7 per-user views in a dedicated schema, a SELECT-only PostgreSQL role with no grants on application tables, a read-only transaction envelope with statement timeouts and row caps, and AST validation of every generated query.
- Built a bounded streaming agent loop over a plain-text protocol instead of native tool calling, keeping free-tier models eligible — with per-user provider choice (Gemini, Anthropic), Fernet-encrypted keys, SSE to web and Android, and 51 tests pinning view isolation and the SQL sandbox.

Tools: Python, FastAPI, SQLAlchemy 2, Alembic, PostgreSQL 16, sqlglot, Gemini & Anthropic APIs, SSE, React 18, TypeScript, Expo, Docker, Neon, Render, Vercel

Skills

- **Programming Languages:** Scala, Python, Java
- **Frameworks & Libraries:** FastAPI, Flask, Spring Boot, Akka, PySpark
- **DevOps & Version Control:** Git, Bitbucket, GitLab, Docker, Kubernetes, Helm, Jenkins, Bazel
- **Databases:** PostgreSQL, Cassandra, MySQL
- **Monitoring & Observability:** Prometheus, Grafana, Elastic APM, JFR, pg_stat_statements
- **Testing & Code Quality:** Gatling, Locust, SonarQube
- **AI & Real-time:** LLM agent loops, Gemini & Anthropic APIs, SSE streaming, Claude Code, Langflow, OpenAI Realtime, Twilio
- **Cloud & Mobile:** Azure, Flutter, Android Studio, Expo Go

Education

• National School of Computer Science (ENSI)

Sept 2019 - June 2022

Engineering Degree in Computer Science ([Accredited by ASIIN with the EUR-ACE label](#))

Tunis, Tunisia

- Grade: 16.53/20 (Highest Honors)

• Preparatory Institute for Engineering Studies of Sfax (IPEIS)

Sept 2017 - June 2019

Pre-Engineering Degree

Sfax, Tunisia

Honors and Awards

- **Promoted to Software Engineer II** – Advanced to the role within one year at COGNIRA, 2023.
- **Technical Interviewer** – Actively participated in conducting technical interviews at COGNIRA, 2023.
- **Top 5% Academic Distinction** – National School of Computer Science (ENSI), 2021.
- **Ranked Top 7% Nationwide** – National Engineering Entrance Exam (Concours Nationaux d'Entrée aux Cycles d'Ingénieurs), Tunisia, 2019.
- **Top Student Award** – Ranked 1st in class during Baccalaureate year, Tunisia, 2017.

Languages

- **English:** Advanced
- **French:** Intermediate
- **German (A1):** Beginner
- **Arabic:** Native